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Learning Objective Statements: Volatility Indexes and VIX Complex



6.3 Volatility Indexes and VIX Complex

Learning Objective Statements: Volatility Indexes and VIX Complex

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<https://skmarketinsights.com>



Learning Objective Statements

- Examine the term structure shapes of backwardation and contango
- Interpret what a shift in the VIX futures term structure shape from contango to backwardation suggests
- Explain how the shape of the VIX futures term structure and the intermonth spreads can provide early warning signals for deploying risk management tools against long SPX positions
- Point out two methods for managing downside risk in the S&P 500®
- Contrast using VIX calls with using S&P 500 puts to manage downside risk
- State the key objective of risk management strategies

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6.3 Volatility Indexes and VIX Complex

Background

VIX is an index and not a tradeable product. It provides a 30-day forward view of volatility. To enable trading in volatility, and for market participants to express a forward view of volatility greater than 30 days, Cboe established VIX futures contracts of different maturities in March 2004.

Weekly and monthly contracts are available, with monthly contracts settling on the Wednesday that is 30 days prior to the third Friday of the calendar month immediately following the month in which the contract expires. Nine monthly contracts are quoted on the Cboe Futures Exchange at www.cboe.com. Six weekly contracts are also available, and they expire every Wednesday.

Besides the VIX Index, Cboe offers VIX9D, VIX3M, VIX6M and VIX1Y indexes for nine days, three months, six months and one year, respectively. All futures contracts are optionable, with the options set to expire on the same day as the futures. It is important to keep in mind that, at expiry, the first-month futures contract converges with and becomes the VIX Index.

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6.3 Volatility Indexes and VIX Complex

Term Structure of VIX Futures and Early Warning Signals

The price of VIX futures, for all maturities, is set by supply and demand in the market. As the time to maturity increases, futures prices could be higher (contango) or lower (backwardation) in value. These two structures, contango and backwardation, are illustrated for two chosen dates in Figure 6.3.1.

The x-axis of Figure 6.3.1 represents monthly futures with increasing maturity dates, labeled F1 through F7. These values are shown in Figure 6.3.2 and represented on the y-axis of Figure 6.3.1.

The upward sloping curve (contango) in red for February 19, 2020, preceding the onset of COVID-19, implies that market participants are quite relaxed about volatility as measured by VIX futures expiring in 30 days. However, as time to maturity increases, uncertainty over future volatility also increases, implying a slight lack of conviction over the continued prevalence of low volatility conditions going forward.

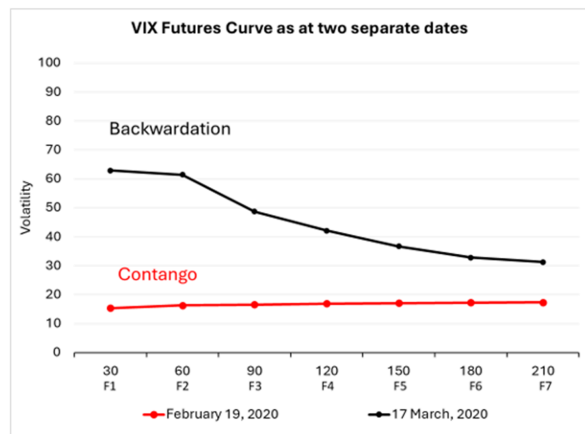


FIGURE 6.3.1 Two VIX futures curves, one in backwardation and one in contango

VIX Futures							
Date (2020)	F1	F2	F3	F4	F5	F6	F7
Feb-19	15.38	16.33	16.53	16.93	17.13	17.23	17.3
Mar-17	68.83	61.43	48.7	42.08	36.7	32.85	31.2

Term Structure of VIX Futures and Early Warning Signals



of the S&P 500 (SPX). Prices had been in an uptrend until early February, with the VIX Index below 15. February 19 is marked by the first blue arrow.

Circumstances changed dramatically in mid-February 2020 as the magnitude of COVID-19's impact became apparent to market participants. SPX declined from around 3400 to 2200 within a month, and volatility increased, as shown by the second blue arrow. The VIX Index rose from below 15 to a high of 82.79 due to this unexpected negative surprise.¹ This caused the VIX futures curve to move in concert and assume a downward sloping (backwardation) structure on March 17, 2020, as shown by the black line in Figure 6.3.1. Market participants had formed a consensus, based on recent events, that extremely high uncertainty existed regarding the future. They expressed this view through higher values of VIX as well as VIX futures up to a maturity of 210 days or seven months.

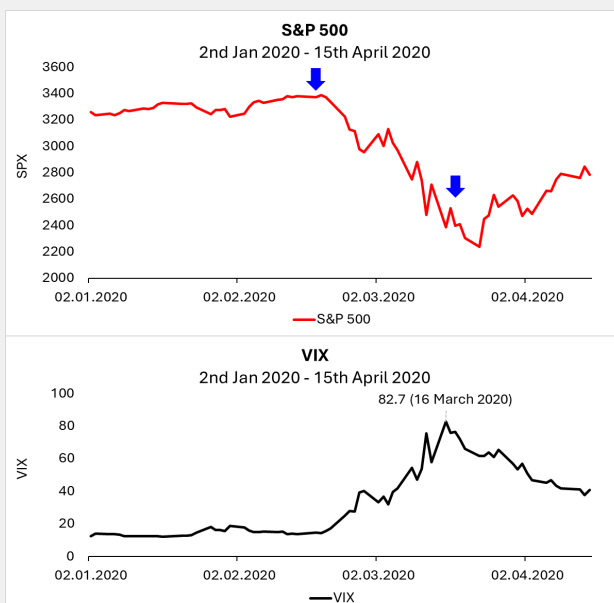


FIGURE 6.3.3 SPX and VIX January 2 through April 15, 2020

This movement in term structure across all maturities can be better visualized using a three-dimensional chart. Figure 6.3.4 is a 3-D chart of VIX futures across the first six maturities during a three-month period, January through March, encompassing the COVID-19 crisis of 2020.

Figure 6.3.4 shows the first six futures contracts with monthly expiries, F1 to F6, plotted along the x-axis, with volatility values of each of these contracts on the y-axis. The movement of these values through time is shown on the z-axis with color coding representing different levels of volatility.

Looking along the z-axis (dd/mm/yyyy), the first-month VIX futures contract (F1) reached a peak value of 68.82 in March 2020, while the outer months (F2) to (F6) peaked at decreasing values. Market participants considered the COVID-19 crisis to be a significant event affecting the market and

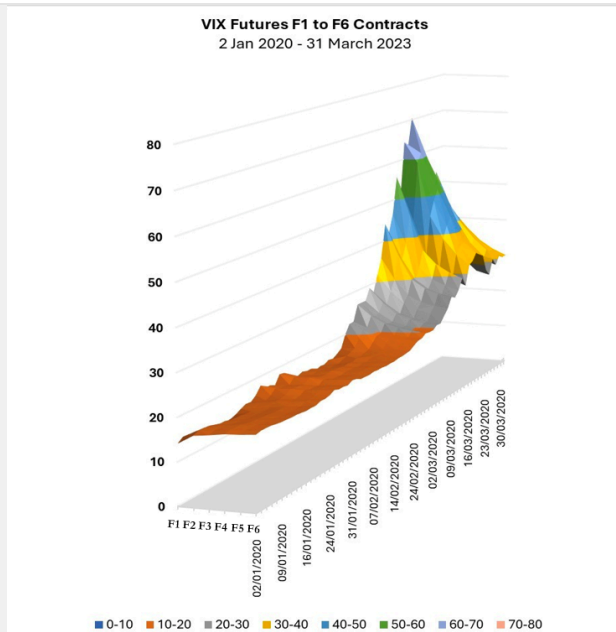


FIGURE 6.3.4 VIX futures January 2 through March 31, 2020

Source: SK Market Insights Ltd.

The first-month futures (F1) rose the most with subsequent maturities rising to a lesser degree. Note that the sixth-month futures contract (F6) remained elevated through the end of March, over its value in early January, implying a high level of nervousness on the part of market participants. This could be attributed to the uncertainty over the timing and resolution of the pandemic during the early stages of the crisis.

Summarizing the observations so far:

- VIX is an index and not tradeable or investable. It expresses a 30-day forward view of volatility.
- VIX futures were established for market participants to take a view of forward volatility greater than 30 days into the future. It represents a time series of forward volatility.
- VIX futures are tradeable and optionable.
- Increasing values of VIX futures with increasing maturities results in an upward sloping, or contango, term structure, normally observable during S&P 500 uptrends.
- S&P 500 bear markets or major corrections are usually characterized by a downward sloping, or backwardated, term structure, with values of VIX futures lower for longer maturities.
- A shift in VIX futures term structure from contango to backwardation occurs when market participants react to an increasing level of perceived risk in the near term.

Deriving Early Warning Signals from VIX Futures Intermonth Spreads

Term Structure of VIX Futures and Early Warning Signals

(backwardation). The slope of the term structure varies between these two extremes, and their gradual shift from one extreme to the other can provide meaningful signals of shifts in risk tolerance, providing a trader with advance notice of changing market conditions. This allows adequate time to deploy risk management tools to minimize downside risk.

One useful way of tracking changes in the VIX futures term structure is through monitoring of intermonth spreads between successive futures of increasing maturities, as explained earlier. Changes to intermonth spreads can be monitored through commercially available trading software or through daily closing data available through www.cboe.com. Historic data is also available from a number of sources, including Cboe.

In the simplest case, tracking changes to the spread between the first- and third-month VIX futures can provide useful indications of upcoming trends and corrections.

Figure 6.3.6 shows this spread for the 2019 through 2024 period, along with the S&P 500 prices. The bottom panel tracking the changes in the F3-F1 spread shows that the VIX futures term structure, as described by this spread, remained mostly in contango during this period. A positive reading indicates that the F3 is greater than F1, meaning the structure is in contango. A significant exception to this occurred in early 2020 due to the COVID-19 crisis. This is evidenced by the spread turning negative, expressing a term structure of backwardation. Apart from this period, the spread remained largely in positive territory, implying no major risk to the downside.

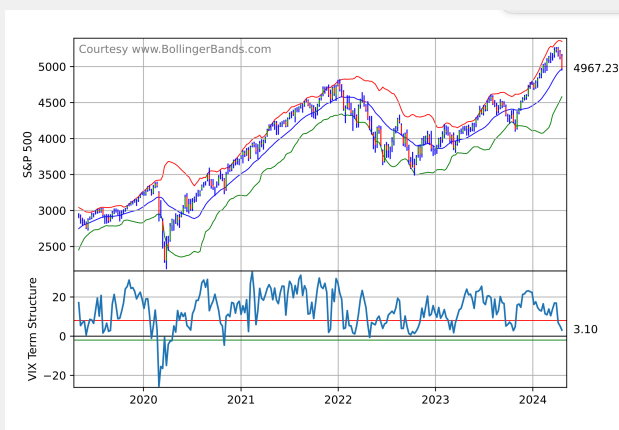


FIGURE 6.3.6 Daily S&P 500 with Bollinger Bands (top panel), spread between first- and third-month futures contracts (bottom panel)

Source: www.bollingerbands.com

Another approach to understanding the slope of the term structure curve is to examine changes in each intermonth spread for the first six months of VIX futures maturity in a color-coded heat map. Upward sloping periods are shown in green while downward sloping periods are shown in red. The wider the spread, the darker the color on the heat map. The

Term Structure of VIX Futures and Early Warning Signals



are interspersed with red (backwardation). The period preceding the Great Financial Crisis in 2008 shows that the intermonth spreads had started to narrow from their average values. This subsequently led to an increase in red with time, implying a shift from an upward (contango) to a downward (backwardation) sloping term structure due to the market sell off in October of 2008. The gradual recovery of the market from this period on into April 2009 is shown in lighter shades of red, returning finally to a green “all clear” shade, implying a return to conditions of lower volatility.

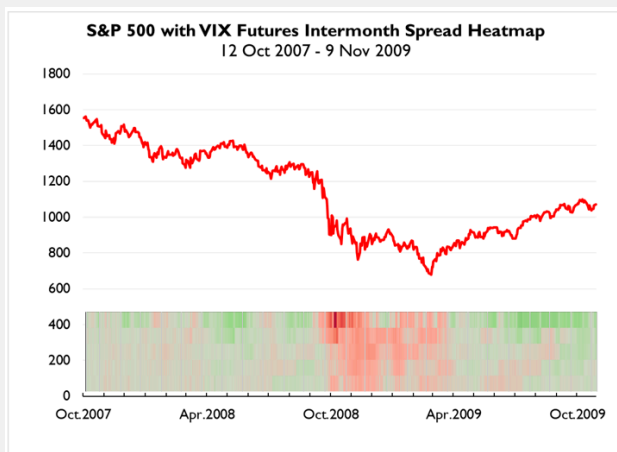


FIGURE 6.3.7 S&P 500 during the bear market of 2007–2009 (top panel), intermonth spreads color coded with backwardation in red and contango in green (bottom panel)

Comparing Figure 6.3.7 with Figure 6.3.8, we see an uptrend in price between 2016 and 2018. There is also a notable difference in the heat map. It shows a continued condition of low volatility through darker shades of green, supportive of market rallies. The dark shades of green imply wider-than-average intermonth spreads due to a firm upward slope.

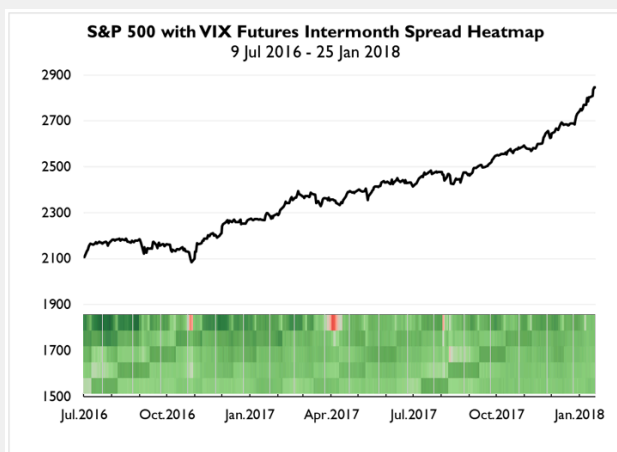


FIGURE 6.3.8 S&P 500 uptrend from 2016 to 2018 (top panel), and intermonth spreads color coded with

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isolated instances of red are observable during November 2016 and April 2017. These represent a negative spread between the first- and second-month futures only, with subsequent maturities retaining a positive spread. The interpretation here is that prevailing risk conditions were estimated to be of short duration and only impacting the first two futures contracts.

In summary, we have seen how the slope of the term structure and changes in intermonth spreads can be useful early warning signals to initiate risk management and minimize downside losses. Next, we will review some available tools for downside risk management.

¹ Highest value of VIX since inception

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S&P 500 Risk Management Tools

VIX Options

VIX calls are the predominant vehicle for risk management purposes, as they are negatively correlated with the S&P 500. This means their options premiums increase during SPX corrections. Additionally, they have a lower risk than VIX futures, as the loss with options is the premium paid to purchase them.

Figure 6.3.9 shows the increasing popularity of VIX calls for “tail-risk” management following the COVID-19 crisis. VIX puts are also available for trading; however, their trading volumes are lower than call volumes. VIX call volumes have been reported to already have had a 54% increase in 2023. Comparatively, VIX puts saw an increase closer to 16% during the same period.²

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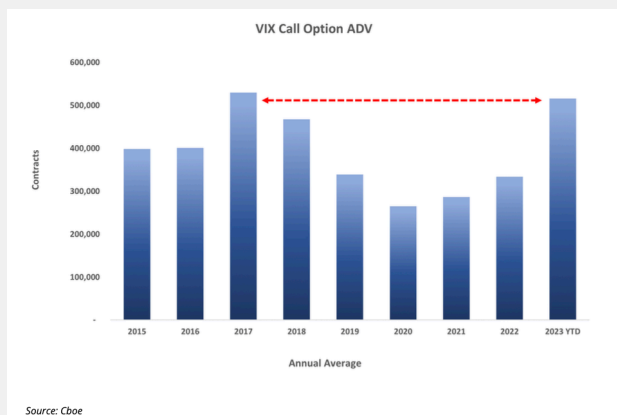
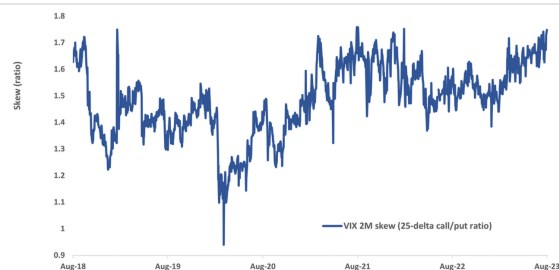


FIGURE 6.3.9 VIX call option average daily volume

The put/call volumes for VIX options work in the opposite way to those of S&P 500 options. Call volumes exceed those of puts during periods of crises. This is shown as a call/put ratio in Figure 6.3.10. It shows the 25-delta³ VIX call/put ratio (referred to here as the VIX 2M skew) between August 2018 and 2023. Since March 2020 (COVID-19 correction), there has been a steady increase in this ratio, levelling off around August 2021. Following a dip in 2022, this ratio has since increased to its all-time high of 1.7, reflecting a strong interest in VIX calls vs. puts.



Source: Cboe

FIGURE 6.3.10 VIX two-month 25-delta call/put ratio

The Cboe Options Exchange has extended its global trading hours and introduced a 24-hour/five-day-a-week trading capability for both VIX and SPX options.⁴ Market participants are now able to trade these options globally nearly 24 hours a day during a five-day work week.

SPX Options

Options on the S&P 500 are also available for risk management. Put options in particular offer downside protection during market corrections, while S&P 500 calls are usually sought after by market participants in anticipation of a market rally. Volatility traders typically look to capture any spread between implied and historical volatility as they appear, capturing the volatility risk premium as discussed in the previous unit. It should be remembered that VIX calls and SPX puts act differently during market corrections.

VIX calls provide tail-risk management against “Black Swan” events, and this largely depends on the extent of VIX movements during a correction. On the other hand, S&P 500 put options are truly a hedge as they move with the underlying without any basis risk; that is, the risk that losses in the investment are not exactly offset by the hedge. They have a fixed strike price, and the option reaches a maximum delta of 1 at expiration when the option is in-the money (i.e., the underlying price goes past the strike price).

Quite often a well-intended hedge may not be effective if SPX’s price continues to rise, and any subsequent correction ends above the put option’s strike price resulting in the option expiring worthless. This can be offset by a process of “rolling” or “rolling up” the strike price to keep the hedge effective. This usually comes at a cost due to premium decay.

The put/call ratio for SPX options is a well-known contrarian indicator for the stock market. When put volumes exceed call volumes, especially at extremes, it provides a measure of the extent of bearishness in the market and is thought to lay the foundation for rallies. Conversely, excessive call volumes vs. put volumes could indicate market tops as bullish enthusiasm peaks.

Recapping the concepts introduced so far:

- VIX futures and options are available as tools for trading and risk management.
- The slope of VIX futures and shifts in the intermonth spreads provide key messages regarding market

diversifying or hedging long equity positions.

² Xu, Mandy. "VIX Options Volume on Pace to Exceed 2017 Record," Cboe.com. August 22, 2023.

<https://www.cboe.com/insights/posts/vix-options-volume-on-pace-to-exceed-2017-record/>.

³ Delta of an option is a measure of the sensitivity of an option's price to changes in the underlying price.

⁴ Moran, Matt. "Trading Strategies for Extended Global Trading Hours for VIX and SPX Options." Cboe.com. November 29, 2021.

<https://www.cboe.com/insights/posts/trading-strategies-for-extended-global-trading-hours-for-vix-and-spx-options/>.

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Power of Convexity in VIX Futures and Options



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Power of Convexity in VIX Futures and Options

S&P 500 options move in relation to the values of the underlying with deep in-the-money options behaving just like SPX.

Figure 6.3.11 captures the change in the S&P 500 Index, the VIX Index, VIX March 2020 futures, and VIX 2020 call options. The price chart for this example was shown earlier in Figure 6.3.3. This chart lets us examine the behavior of VIX futures and options during crises.

Based on closing values, if one were to buy the VIX first-month futures or the VIX March 20 calls, both expiring on March 18, 2020, a decrease in the S&P 500 of 25.3% would have triggered a 428% increase in the VIX Index. A long first-month VIX futures strategy (F1) would have resulted in a profit of 358%, and a corresponding investment in the VIX March 20 calls would have resulted in a profit of 6393%.

The takeaway is that VIX moves, during S&P 500 corrections, are rich in convexity (the multiplier effect) and corresponding moves in near month futures capture around 80-85% of such a change. Convexity here is defined as the sensitivity of the option's price to changes in the underlying asset's price. The movement in VIX calls are even more convex, due to an increase in implied volatility, leading to a 15-fold increase in value, pointing to the power of options.

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Changes in Reported Closing Values Over 20 Days in 2020
(February 19 to March 17, 2020)

Date	S&P 500 Index	VIX Index (not investable)	VIX March 2020 Futures (that expired on March 18 th)	VIX March 20 call options (that expired on March 18 th)
19-Feb-2020	3386.15	14.38	15.39	\$0.75
17-Mar-2020	2529.19	75.91	70.55	\$48.70
% change	-25.3%	428%	358%	6393%

FIGURE 6.3.11 The power of options

Next, Figure 6.3.12 demonstrates the convexity in the VIX Index versus the S&P 500 Index during the February 1990 to August 2019 period.

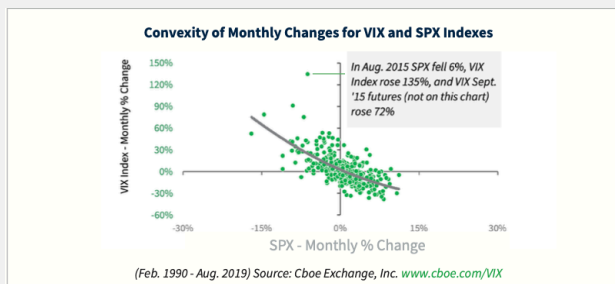


FIGURE 6.3.12 Convexity of SPX and VIX February 1990 through August 2019

A long VIX futures strategy or a VIX call options strategy may not be a panacea for downside risk. While they provide catastrophic loss protection, there are associated costs. In the case of VIX futures, buying and holding a near-month VIX future in a portfolio could be a drag for the simple reason that markets remain in contango between 65% and 75% of the time. As the futures approach expiry, the position needs to be closed and the next month's futures contract needs to be purchased, a term referred to as the "roll." Roll costs during contango periods involve selling futures that have declined in value and purchasing ones at higher prices. Eventually, the roll costs end up having a material impact on the profitability of the portfolio.

This is exemplified by Figure 6.3.13 which shows the ProShares VIX short-term futures ETF (VIXY) plotted in blue and the S&P 500 Index in red. The value of VIXY spikes briefly during market corrections, but it remains in an overall downtrend due to the cost of the roll in contango or upward sloping markets. Hence VIX futures are not advisable for hedging long-term "buy and hold" strategies on SPX, particularly during contango markets. They are, however, useful as a short-term diversifier.

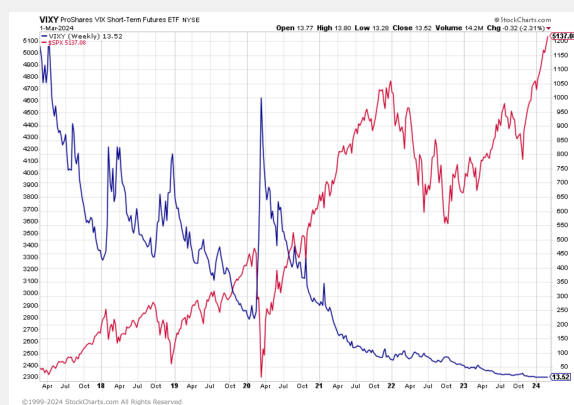


FIGURE 6.3.13 Weekly line chart VIXY in blue and SPX in red

VIX futures, as shown earlier. When judiciously applied, they can provide downside protection. The downside risk is limited to the premium paid for the VIX calls.

Recognizing this feature of VIX options, Cboe has designed a VIX Tail Hedge Index (VXTH) that provides diversification for a portfolio of SPX with one-month 30-delta VIX call options using a formula based on the value of the first-month VIX futures shown in Figure 6.3.14.

F = One Month Forward Value of VIX	X = Portfolio Weight of VIX Calls
F ≤ 15%	0%
15% < F ≤ 30%	1%
30% < F ≤ 50%	½%
F > 50%	0%

FIGURE 6.3.14 Formula for VXTH

Source: Cboe.com/VXTH

If the front month futures trades below 15, no calls are purchased. If the value of this future increases to between 15 and 30, a 1% portfolio weight in one-month 30-delta VIX calls is purchased. These calls are rolled every month upon expiration, meaning new calls with the same characteristics are purchased. When the value of the first-month VIX futures is in the range of 30 to 50, the weight is reduced to 0.5%. Finally, if the futures value exceeds 50, then the VIX calls are liquidated.

This strategy has the benefit of deploying risk management through VIX calls only when the probability of tail-risk increases and it exits when the risk has receded. It has the benefit of mitigating downside risk with a minimum impact on returns. The performance of such a strategy is compared with an SPX buy and hold strategy in Figure 6.3.15.

Metrics for Strategy Indices April 2006				
	Annualized Return	Standard Deviation	Downside Deviation (MAR = 0%)	Maximum Drawdown
VXTH-Cboe	10.0%	17.8%	8.6%	-37.4%
VIX Tail Hedge Index	10.1%	15.5%	10.4%	-51.0%
S&P 500 Index				

FIGURE 6.3.15 VXTH strategy versus buy and hold

Source: Cboe.com/VXTH

hold strategy, but it has a lower drawdown (5.4 vs. 5.7%) and a lower downside deviation (8.6 vs. 10.4%). This example is shown for demonstration purposes only. The reader is encouraged to devise their own trigger for hedging tail risk that could reduce losses further, thereby improving downside protection.

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SPX Put-Based Strategies



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SPX Put-Based Strategies

For comparison purposes, several SPX put strategies that could be applied for risk management, in lieu of VIX options or futures, are presented. Keeping in mind that S&P 500 options strategies have their pros and cons discussed earlier, we examine the following:

1. A simple put option buying strategy that provides downside protection by purchasing a one-month 5% out-of-the-money put option on SPX. When the put option expires, a new put option with a similar strike is purchased. Such an index is labelled "PPUT" and has been created by Cboe with specification details available at www.cboe.com/PPUT
2. A variation on the first strategy is to purchase a 10% out-of-the-money SPX put option that expires on a quarterly cycle. Upon expiration, a fresh quarterly option is purchased with similar characteristics. The performance of such an index is available at www.cboe.com/PPUT3m
3. A build on the simple long put strategy is to finance the long put with a short call. The cost of the long put is partly financed by selling a 10% out-of-the-money SPX call with the same expiration date. Upon expiration of the options, the same strategy is repeated with forward puts and calls. Details of this strategy are available at www.cboe.com/CLI
4. A variation on the above strategy is to purchase a 5% out-of-the-money SPX put that expires on a quarterly cycle. The cost of the long put is partly financed by selling a 10% out-of-the-money SPX call that expires on a quarterly cycle. Upon expiration, fresh quarterly put and call options are purchased with similar characteristics. The performance of such an index is available at www.cboe.com/CLL3M
5. Lastly, a zero-cost collar strategy is devised through a put vertical spread, buying a 2.5% put and selling a 5% put. The lower premium on this spread is defrayed by selling a call option with the same expiry, above the existing SPX index value. This strategy is repeated every month at expiry with a fresh set of puts and calls. Details of this strategy can be found at www.cboe.com/CLLZ

A comparison of the above five strategies against the SPX buy and hold strategy, between April 2006 and February 2024, is shown in Figure 6.3.16 along with the VXTH metrics.

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	Annualized Return	Standard Deviation	Downside Deviation (MAR = 0%)	Maxi Draw
VXTH-Cboe VIX Tail Hedge Index	10.0%	17.8%	8.6%	-37
PPUT-Cboe S&P 500 5% Put Protection Index	7.5%	12.1%	7.9%	-38
PPUT3M-Cboe S&P 500 Tail Risk Index	7.9%	12.2%	7.8%	-37
CLL-Cboe S&P 500 95-110 Collar Index	6.4%	10.7%	6.9%	-35
CLL3M - Cboe S&P 500 3-Month Collar Index	6.0%	9.8%	6.5%	-28
CLLZ - Cboe S&P 500 Zero-Cost Put Spread Collar	6.5%	11.9%	8.4%	-43
S&P 500 Index	10.1%	15.5%	10.4%	-51

FIGURE 6.3.16 SPX put-based strategies versus SPX buy and hold

Source: [Cboe.com](https://www.cboe.com)

All SPX put strategies have a lower annualized return than the S&P 500 buy and hold. Their returns are also lower than the VXTH strategy (6 to 7.9 vs. 10%). However, they have lower standard deviations (9.8 to 12.2 vs. 15.5%) as well as downside deviations than the SPX buy and hold (6.5 to 8.6 vs. 15.5%).

The downside deviation as well as the maximum drawdown of the three-month collar strategy (CLL3M) (6.5% and 28.2%) is the lowest of all SPX put strategies for the period examined. All SPX put strategies have lower drawdowns than the SPX buy and hold. All strategies have a lower Beta than the SPX buy and hold and slightly lower Sharpe and Sortino ratios.

Of all the strategies reviewed, the VXTH tail-hedge strategy provides the best annualized return while the three-month SPX collar strategy provides the best downside protection during the period shown.

The above examples demonstrate various possibilities to manage tail risk using VIX futures, VIX options, and SPX options.

The key to risk management strategies is to limit downside risk while giving up minimal upside returns.

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